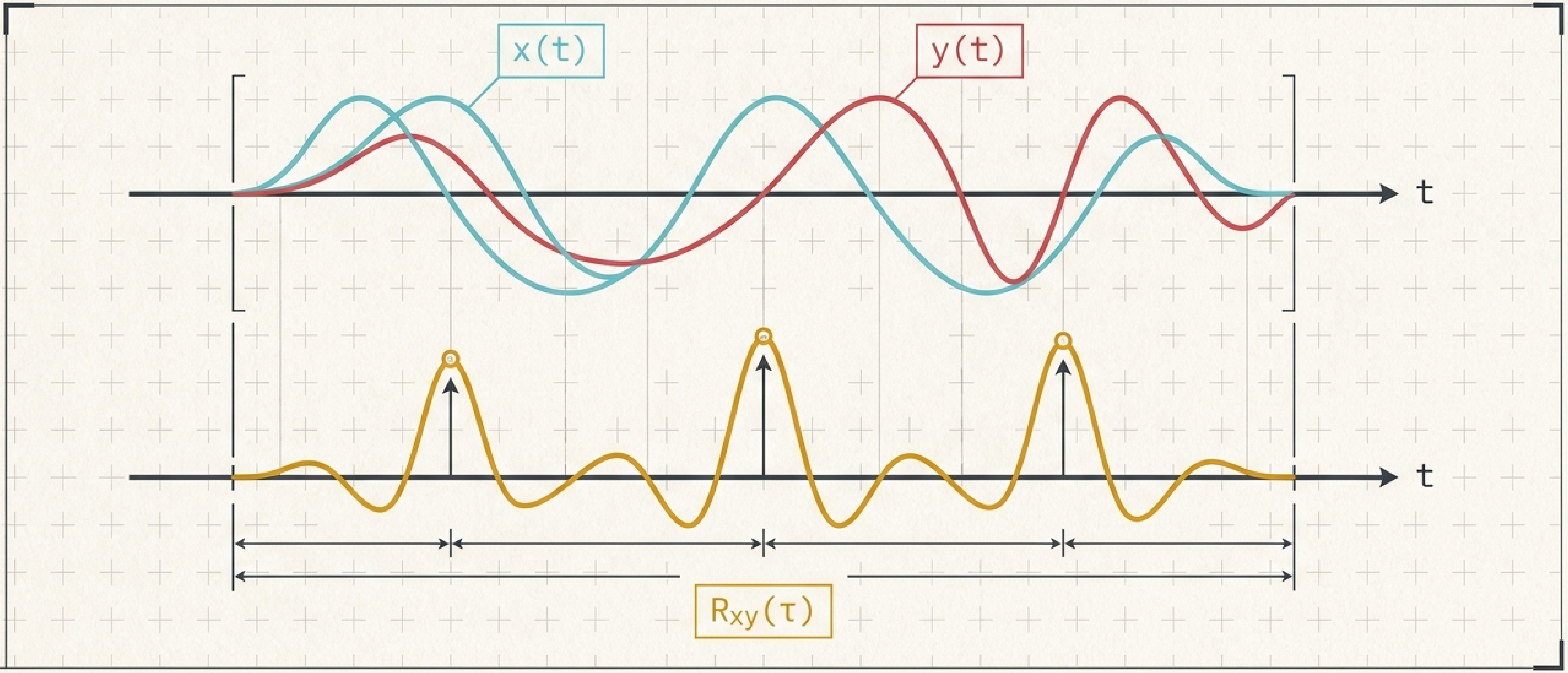
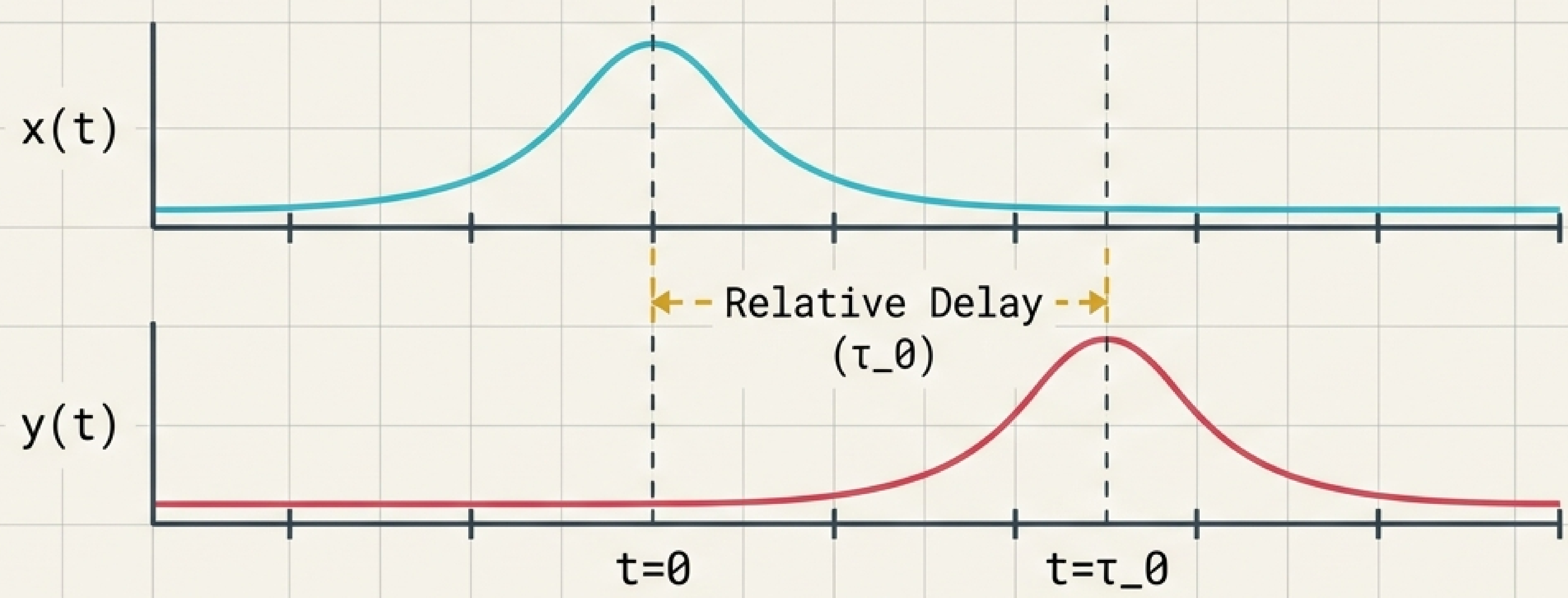
Correlation of Signals

Measuring Similarity in Time



The Core Problem: Measuring Delay-Dependent Similarity

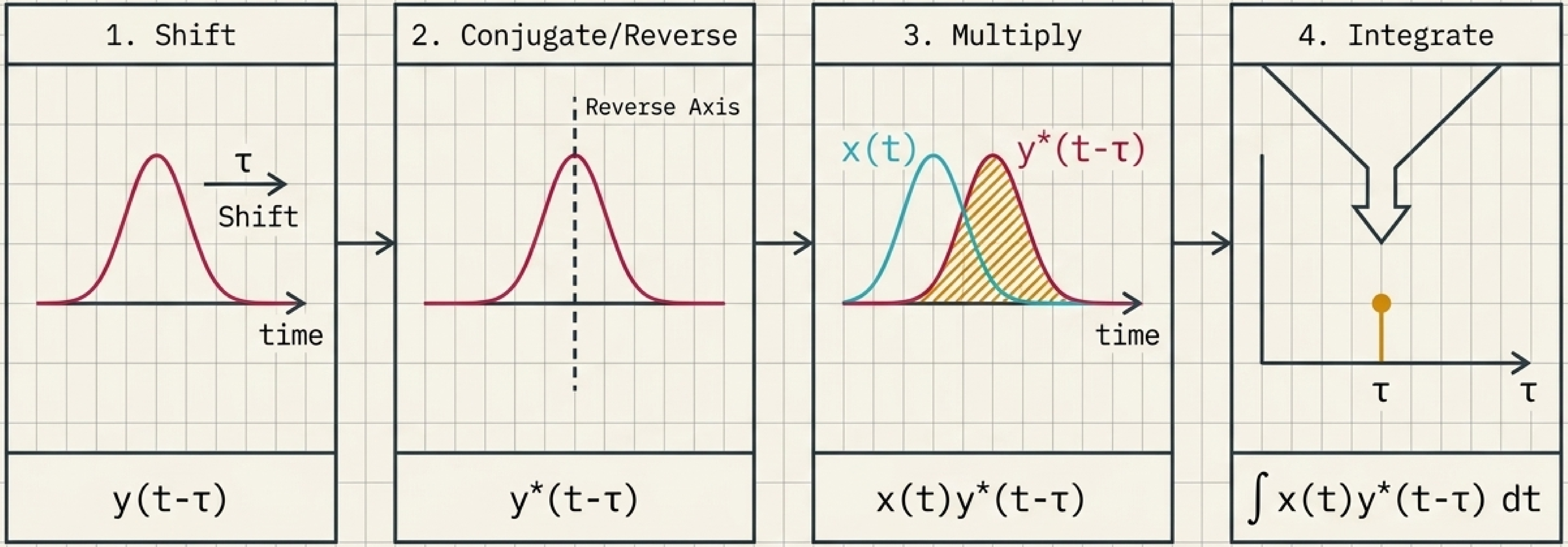
If $y(t)$ resembles a delayed version of $x(t)$ such that $y(t) \approx x(t-\tau_0)$, how do we mathematically prove it? Correlation answers this by quantifying similarity as a continuous function of relative displacement.



Correlation measures similarity versus time shift.

The Engine of Correlation: A 4-Step Pipeline

To find similarity at any given delay τ, we perform a sliding inner product between the reference signal and the shifted target.



Deconstructing the Cross-Correlation Integral

Correlation is fundamentally an inner product calculated after a relative shift. A large positive value indicates similar signed variations; near zero indicates weak linear similarity.

The Output
The resulting similarity score at delay τ.

$$R_{xy}(\tau) = \int_{-\infty}^{+\infty} x(t)\, y^*(t-\tau)\, dt$$

Integration
Summing the overlapping area.

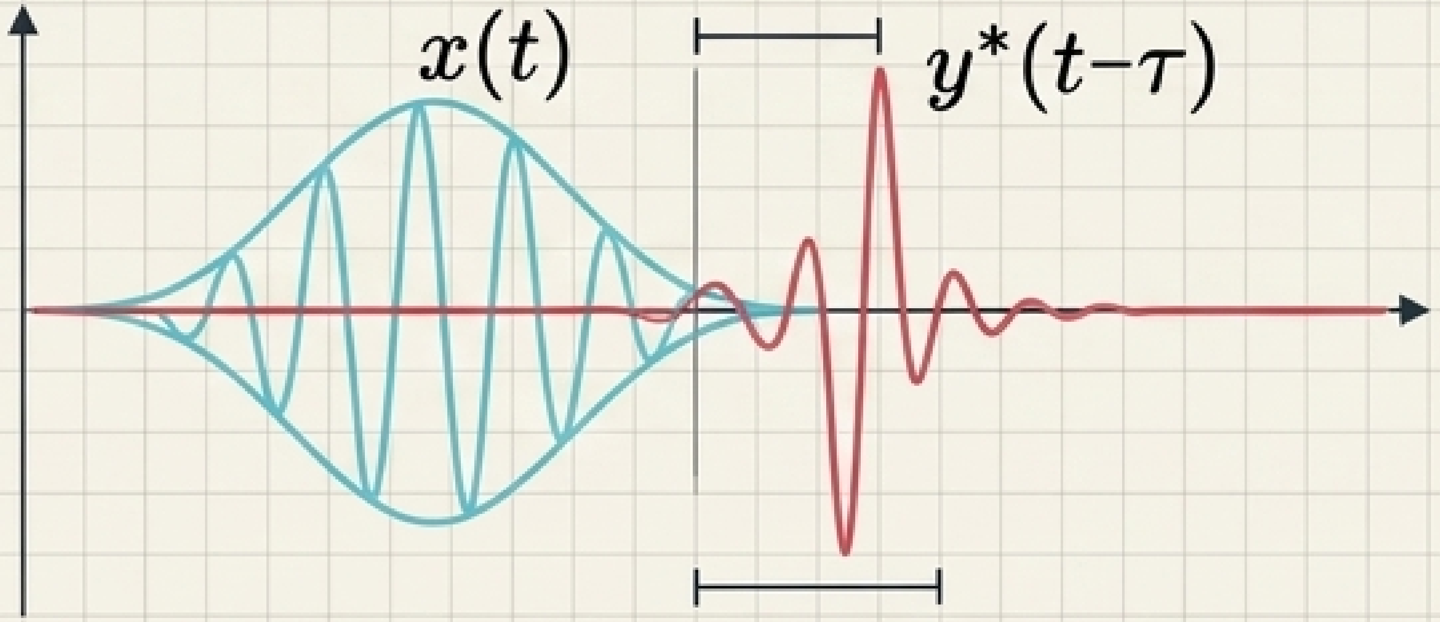
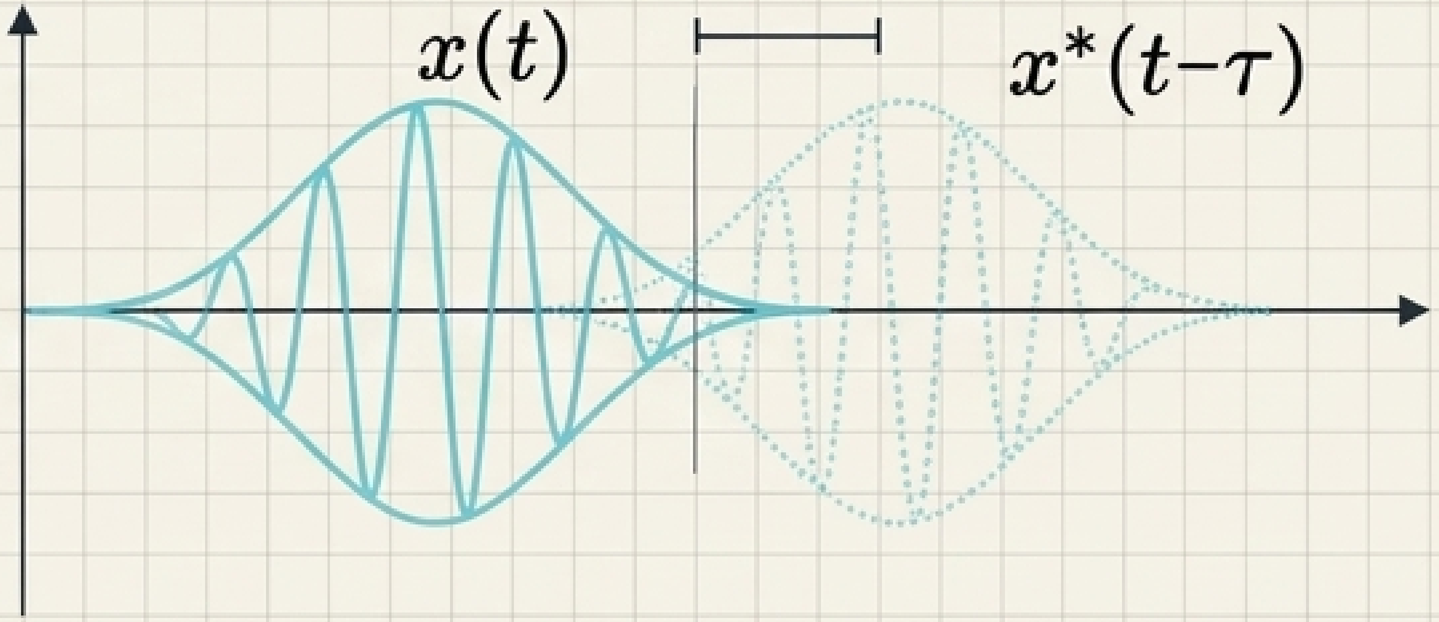
Reference
The static reference signal.

Target
The target signal (complex conjugated).

Sliding Window
The shifting mechanism.

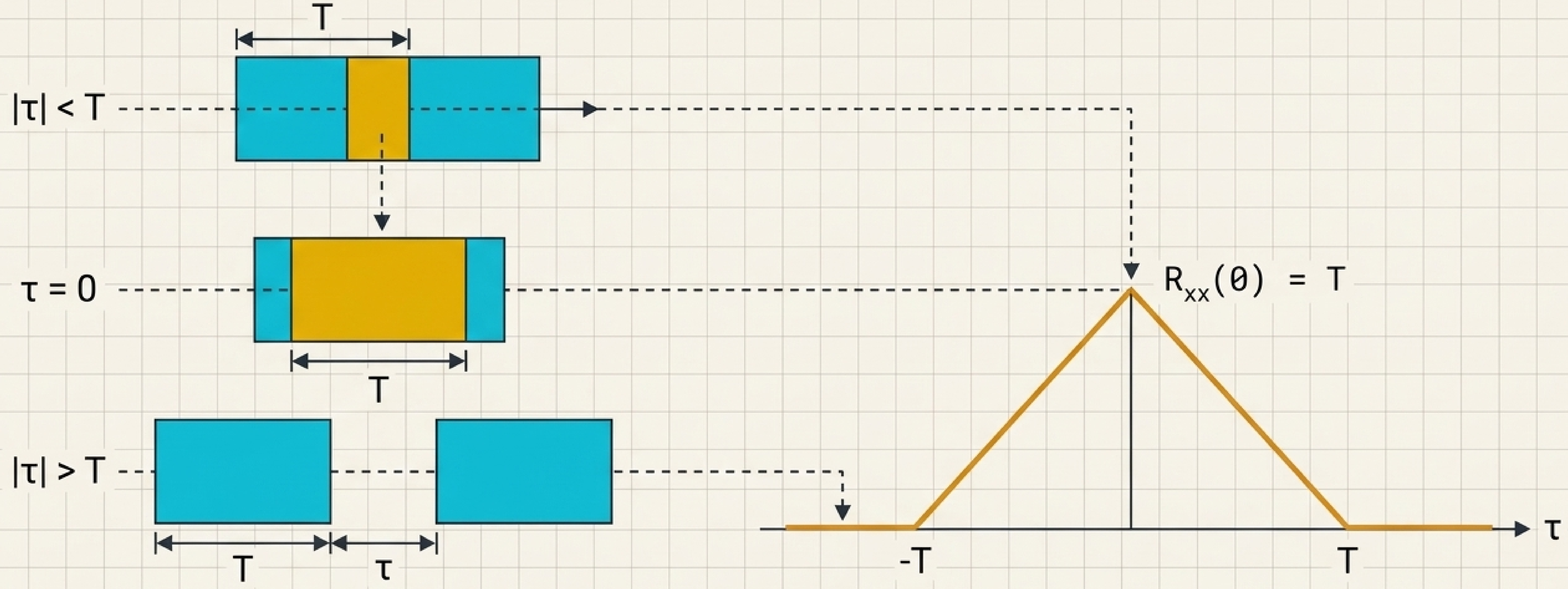
The Fundamental Divide: Two Modes of Measurement

While the mechanics remain identical, substituting a second signal for a delayed copy of the first radically alters the function's physical meaning.

Cross-Correlation Finding Others	**Autocorrelation** Finding Self
	
$R_{xy}(\tau) = \int x(t)y^*(t-\tau)dt$	$R_{xx}(\tau) = \int x(t)x^*(t-\tau)dt$
Radar, Sonar, Time-delay estimation	Periodicity, repeated patterns, hidden structures

Visualizing Overlap: The Rectangular Pulse

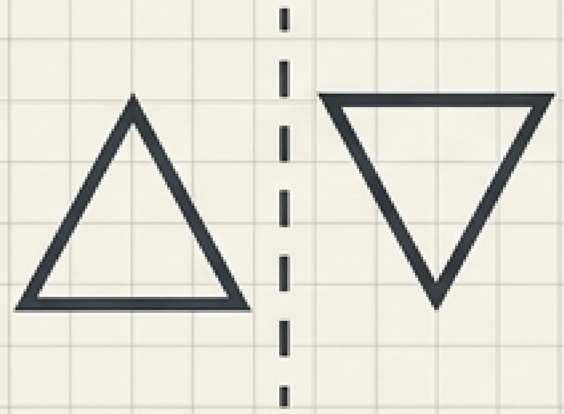
For a rectangular pulse where $x(t)=1$ for $|t| \le T/2$, the autocorrelation is determined purely by geometric intersection. The overlap length is $T-|\tau|$, yielding a triangular function that peaks perfectly at zero lag.



The Immutable Laws of Correlation

Regardless of the signal shape, all correlation functions must obey these strict mathematical constraints derived from inner product properties.

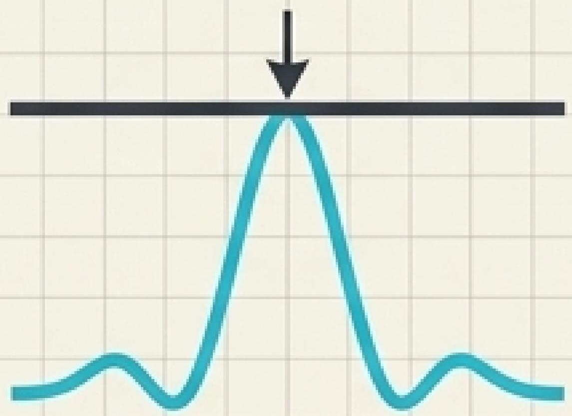
Symmetry



$$R_{xx}(\tau) = R^*_{xx}(-\tau)$$

Conjugate Symmetry. For real signals, autocorrelation is a perfectly even function.

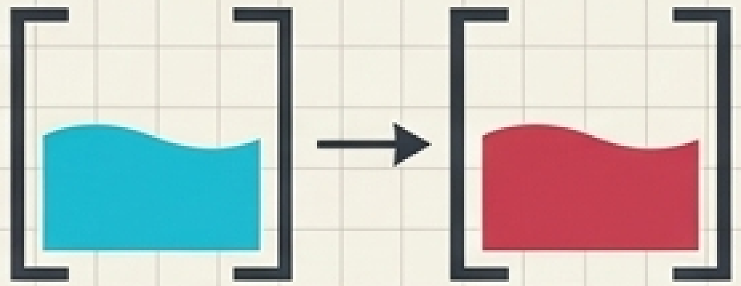
The Absolute Ceiling



$$|R_{xx}(\tau)| \leq R_{xx}(0)$$

A signal can never be more similar to anything else than **it is to itself at zero delay.**

Cauchy-Schwarz Limits



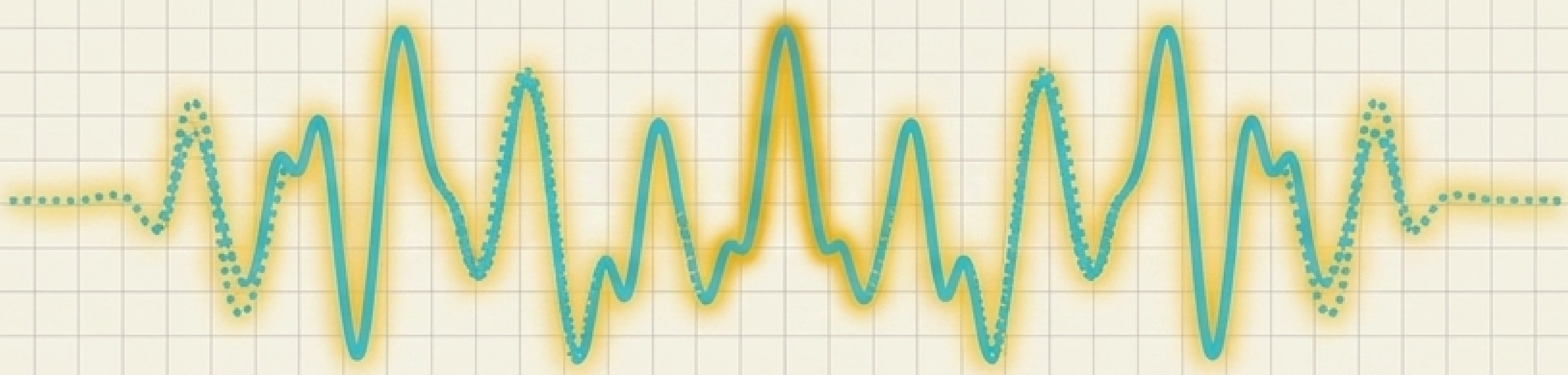
$$|R_{xy}(\tau)| \leq \sqrt{E_x E_y}$$

Energy Limits. Cross-correlation magnitude is strictly bounded by the total energy of both signals.

The Peak at Zero: Autocorrelation and Energy

When $\tau = 0$, the temporal shift is eliminated. The operation reduces to integrating the squared magnitude of the signal, meaning the maximum self-similarity of a finite-energy signal is exactly its total energy.

$$R_{xx}(0) = \int x(t)x^*(t)dt \rightarrow x(t)x^*(t) \rightarrow |x(t)|^2 \rightarrow \int |x(t)|^2\, dt = E_x$$



Maximum Self-Similarity = Total Energy

The Cognitive Trap: Correlation vs. Convolution

They look mathematically identical, but serve entirely different physical purposes. Convolution determines an output through an LTI system; correlation measures structural alignment.

Correlation	**Convolution**
Similarity Operation	System Response
$\int x(t)\, y^*(t-\tau)\, dt$	$\int x(\tau)\, h(t-\tau)\, d\tau$

The Mathematical Link

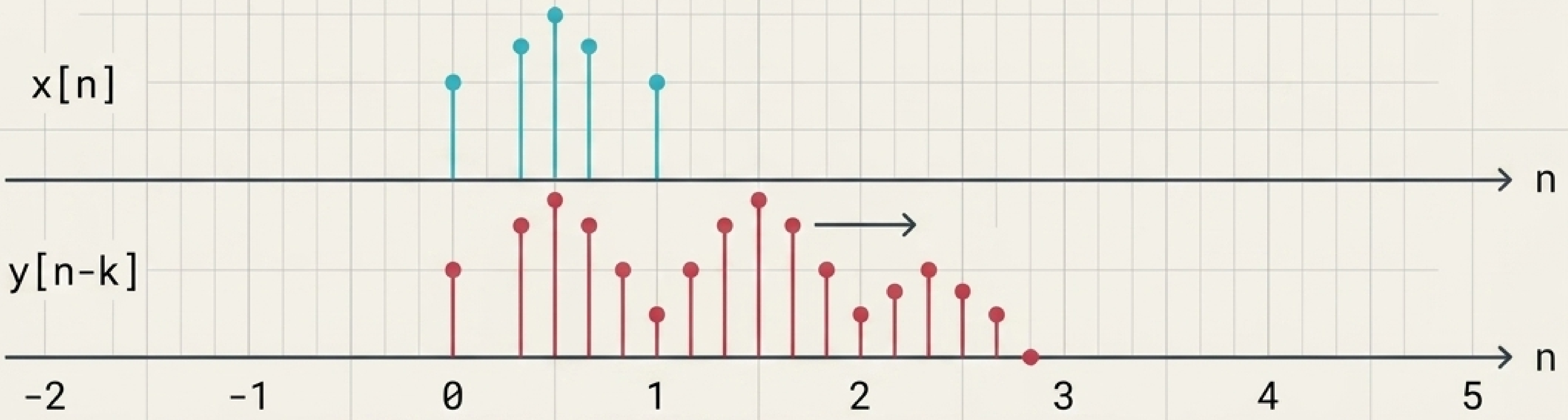
$$\boxed{x(t) \star y(t) = x(t) * y^*(-t)}$$

Correlation is simply convolution with a reversed and conjugated signal.

The Discrete World: Integrals to Summations

$$R_{xy}[k] = \sum_{n=-\infty}^{+\infty} x[n]\, y^*[n-k]$$

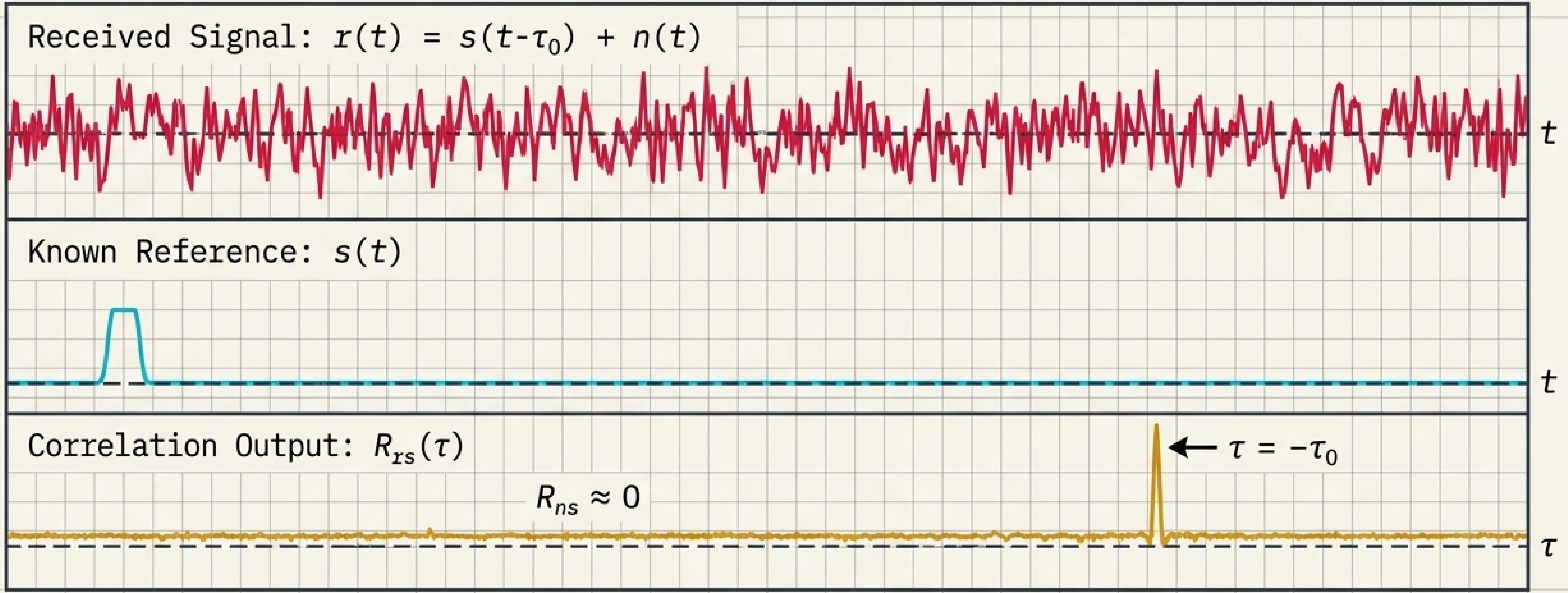
In digital signal processing, continuous integration becomes discrete summation. The continuous delay τ is replaced by integer sample shifts k, but the geometric principles of overlap remain identical.



Shift k = 0	Overlapping Samples: 3	Output: 3
Shift k = 1	Overlapping Samples: 2	Output: 2
Shift k = 2	Overlapping Samples: 1	Output: 1

Detecting the Invisible: Signals in Noise

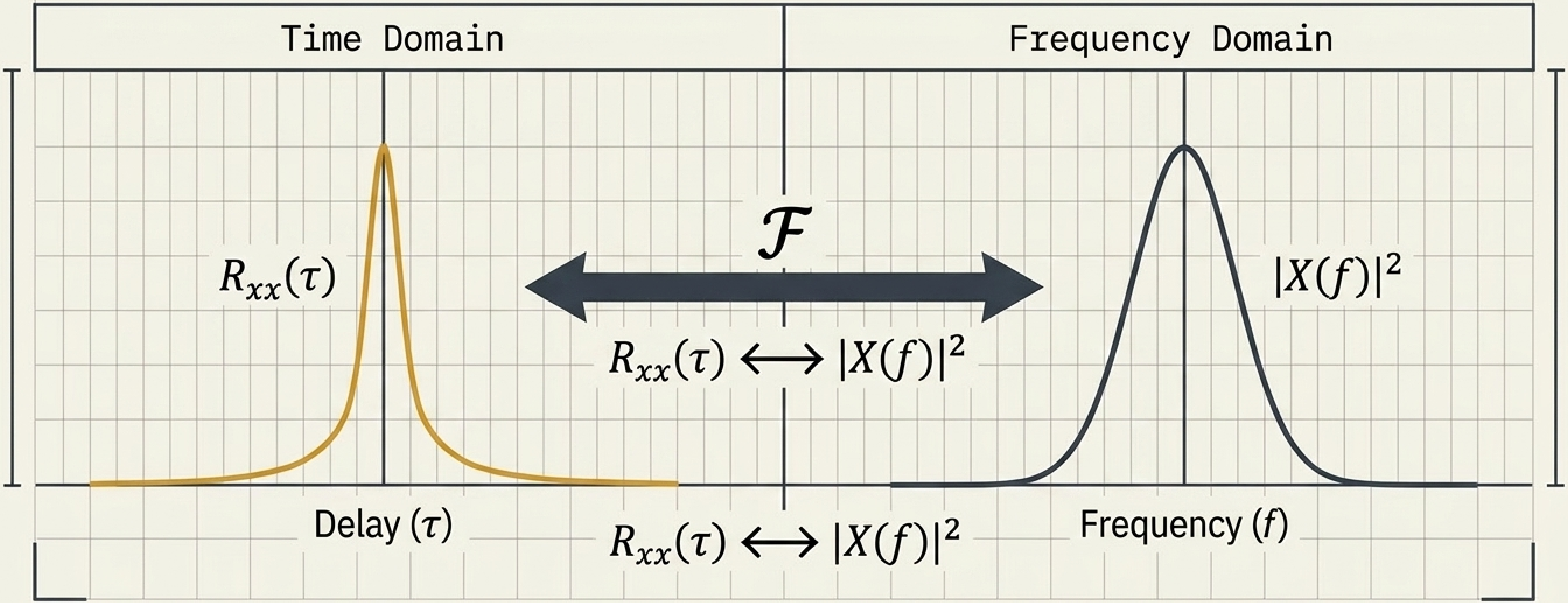
If noise is uncorrelated with a reference signal, correlation filters it out. Searching for the strongest peak forms the basis of matched-filter detection in radar and communications.



Note: Sign convention matters. Peak occurs at $\tau = -\tau_0$ under this specific integral definition.

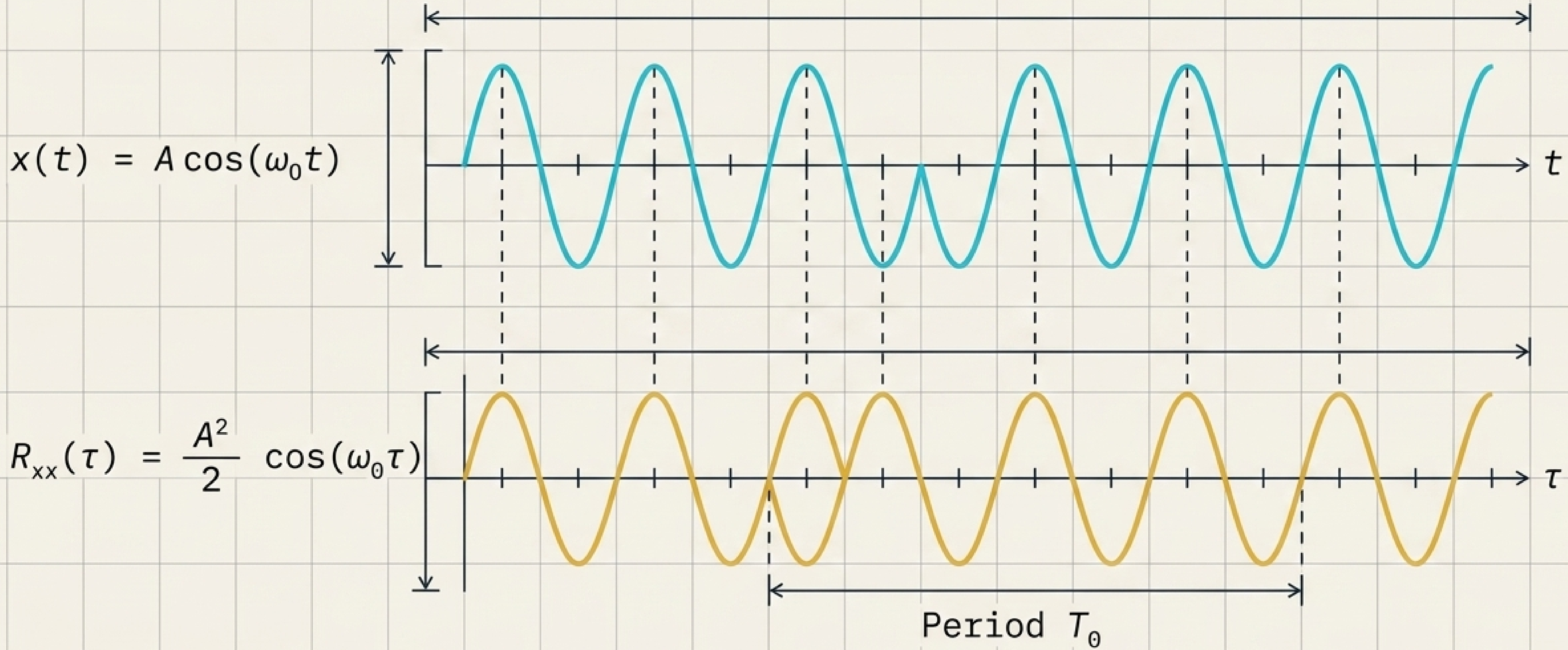
The Time-Frequency Bridge: Wiener–Khinchin Theorem

Autocorrelation and spectral energy density are mathematical mirrors. The structural similarity of a signal over time contains the exact same information as its energy distribution over frequency.



Finding the Pulse: Periodic Autocorrelation

A pure sinusoid correlates infinitely with itself at periodic intervals. For any periodic signal, $R_{xx}(\tau + T_0) = R_{xx}(\tau)$, making autocorrelation the ultimate tool for estimating unknown periodicities.



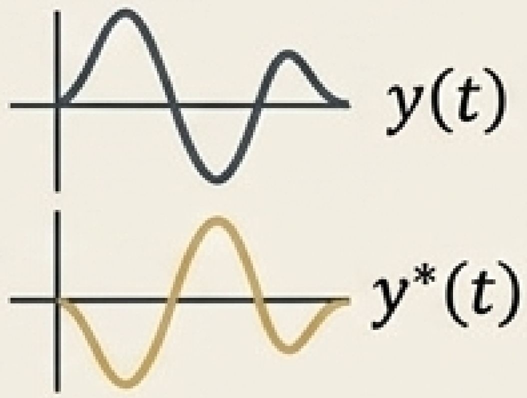
Gemini Notebook

Diagnostic Table: 6 Common Correlation Errors

Ensure accurate signal analysis by avoiding these fundamental theoretical and mathematical traps.

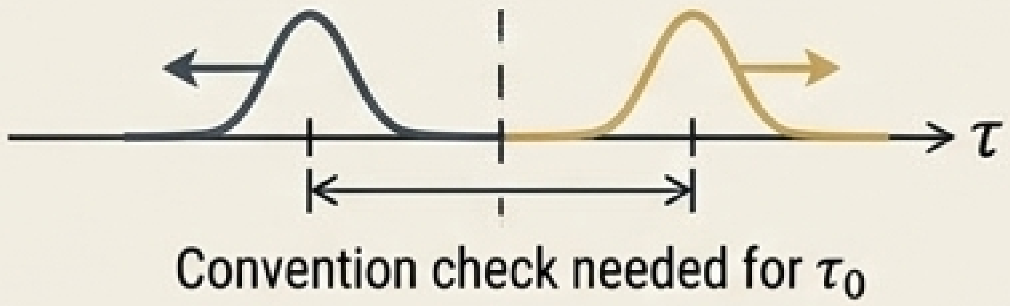
Conjugation

Forgetting $y^*(t)$. The complex conjugate is essential for complex signals.



Delay Sign

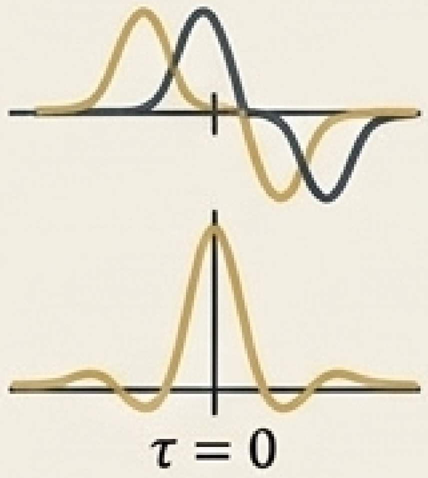
Assuming τ_0 direction. Always check the textbook's shift convention before interpreting direction.



Peak at Zero

Assuming zero-delay peak. Cross-correlation does NOT have to peak at zero.

Only Autocorrelation is bound to $R_{xx}(0)$.

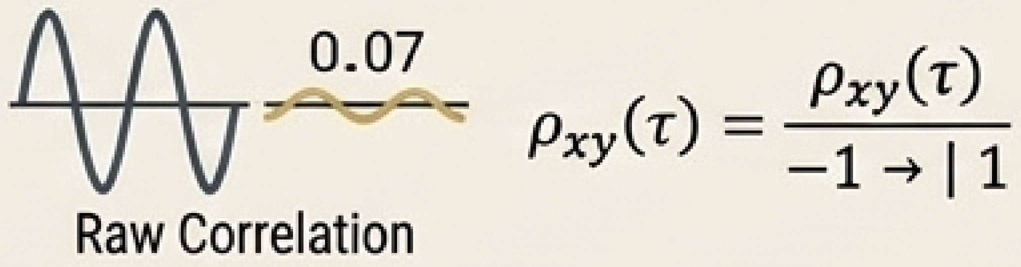


Causation

Misinterpreting meaning. Large correlation indicates similarity or association, never causal relationships.

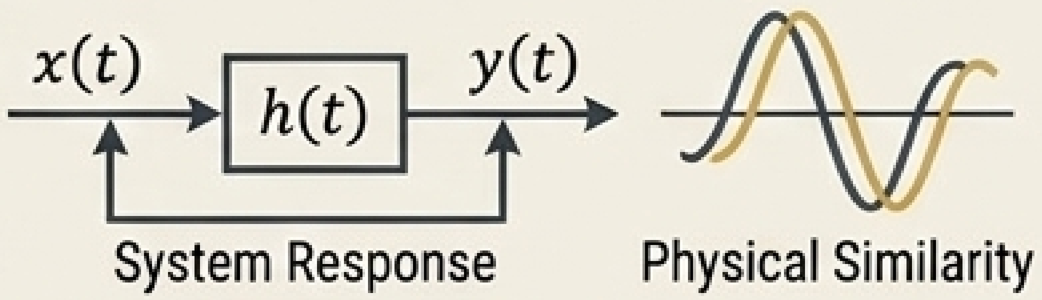
Signal Energy

Ignoring amplitude. High raw correlation might just mean massive signal energy. Use Normalized $\rho_{xy}(\tau)$.



Convolution

Confusing the math. One finds a system response, the other finds physical similarity.



The Master Blueprint: Correlation Reference

Shift → Multiply → Integrate → Measure Similarity

Definitions	Symmetry

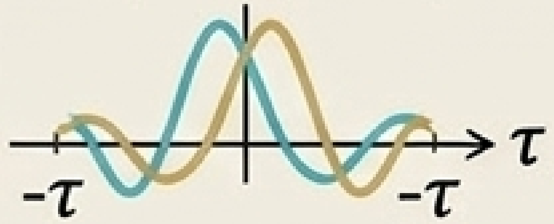
Definitions

Continuous: $R_{xy}(\tau) = \int \underbrace{x(t)}_{x(t)}\,\underbrace{y^*(t-\tau)}_{y^*(t-\tau)}\,\underbrace{dt}_{dt}$

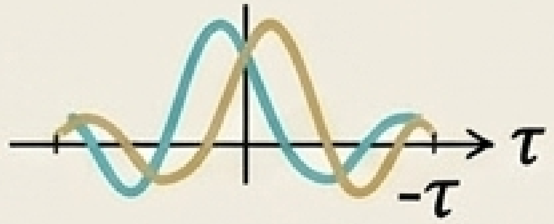
Discrete: $R_{xy}[k] = \sum_{x[n]} \underbrace{x[n]}_{x[n]}\underbrace{y^*[n-k]}_{y^*[n-k]}$

Symmetry

Auto: $R_{xx}(\tau) = R_{xx}^*(-\tau)$



Cross: $R_{xy}(\tau) = R_{yx}^*(-\tau)$

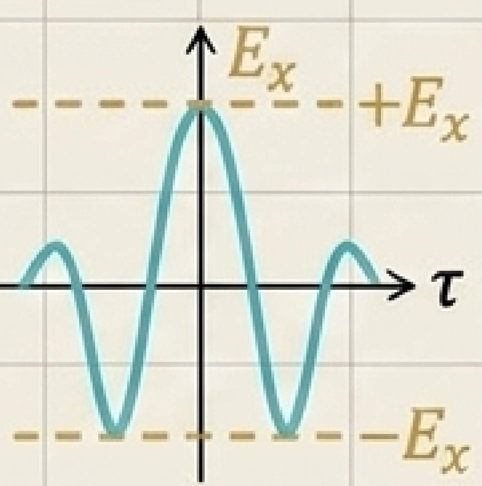


Bounds & Energy

Maximum at Zero: $R_{xx}(0) = E_x$

Absolute Ceiling: $|R_{xx}(\tau)| \leq R_{xx}(0)$

Cauchy-Schwarz: $|R_{xy}(\tau)| \leq \sqrt{E_x E_y}$



Transform Theorems

Cross-Correlation: $R_{xy}(\tau) \leftrightarrow X(f)Y^*(f)$

Wiener-Khinchin: $S_{xx}(f) = \mathcal{F}\{R_{xx}(\tau)\}$